Name: Team Timbuktu **Date:** 2/24/25

Element D: Design concept generation, analysis, and selection

Process for generating & comparing possible solutions. The process for generating possible solutions

Design solution choice: The design solution was to recreate a smaller crane version. The proposed miniature crane system for the eland toy hanger would function similarly to a traditional crane, scaled down for zoo use. The crane's base, embedded in a concrete foundation below ground, provides stability like industrial cranes anchored to heavy platforms. A beam planted into the ground would support a horizontal arm equipped with a pulley and rope system. Users would operate the toy hanger via a lever or hand to lower the rope into the eland exhibit, allowing the toy to be hung.

Process of Generating & Comparing Solutions: We started by thinking of ideas. We looked at big cranes that builders use and tried to make them smaller for the zoo. We had to remember special zoo rules about keeping animals and workers safe. First, we learned how real cranes stay steady and hold heavy things. Then we thought of different ways to make our toy hanger. Some ideas used motors, some just used ropes and wheels. We drew pictures and made small models to see how our ideas might work. We wanted something easy for the zookeepers to use but strong enough to last outside. We also had to make sure it didn't cost too much money. After looking at all our ideas, we picked the best ones to try.

Plan of Action: The plan of action was a scoring matrix developed to evaluate concepts against critical criteria. We used a scale of 1 to 5 to prioritize crucial factors for our

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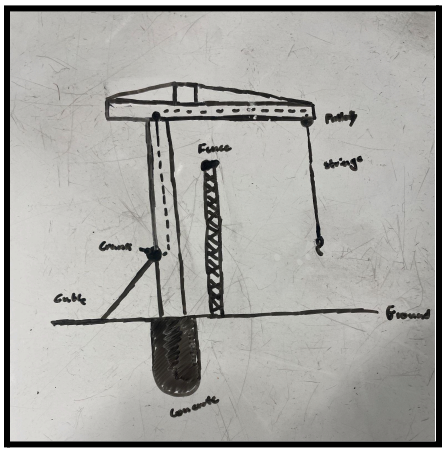
crane. Safety was prioritized first, leading to the elimination of motorized systems due to risks and cost. Durability assessments favored robust materials like steel beams and concrete foundations, but we went for wood as it matched our budget. User engagement became a deciding factor, with manual operation methods, such as levers or hand cranks, scoring higher than automated systems, as they encouraged direct interaction.

Names	P1. Height	P2. Strength	P3.Safety	P4. Accessibility	P5. Longevity
Garret	1	4	5	3	2
Miles	2	3	5	4	1
Brandon	2	5	4	3	1

Design Prototype & Final Design: The final selected solution, a manually operated miniature crane, emerged as the optimal choice. Its design mimics industrial cranes but scales down components for zoo use. The system features a horizontal arm supported by a vertical beam anchored in a concrete base. A pulley and nylon cords allow users to lower a hook into the exhibit, combining mechanical functionality with hands-on engagement. This approach addressed safety through the hand operation, durability, and adaptability.